

# ROLL CENTER TUNING GUIDE

WAGGONER DESIGNS RACING ENGINEER · B6 / B74 FAMILY NUMBERS

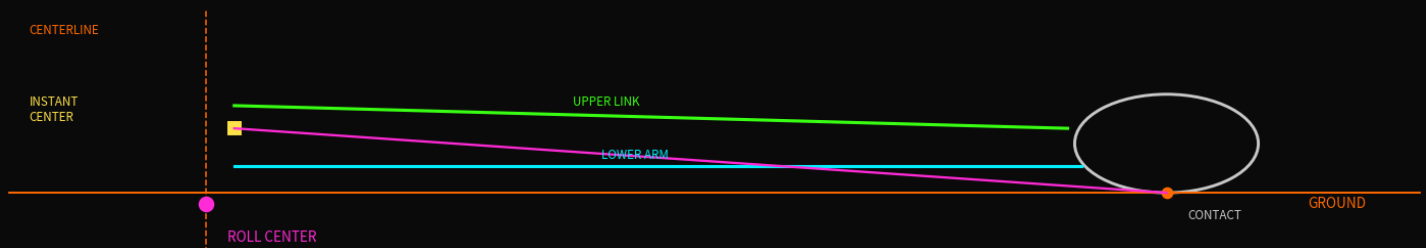
Roll center height is one of the strongest chassis levers you have, and it is easy to mix up with spring rate. Every inner pill, ball stud, axle insert, and ride-height click moves that point a different amount. The tables on page 2 are for the Associated B6.2 and B74 rear ends from a geometry model of those kits.

What it is. The roll center is where the tire's cornering force is treated as hitting the chassis. The center of gravity is where the car's mass wants to fly wide. The gap between those two points is the lever that rolls the chassis. Low roll center = more roll, the end feels softer, usually more grip. High roll center = less roll, the end feels stiffer, usually less grip. If roll center sits at the same height as CG, that end will not lean. Off-road buggies often run the roll center below the dirt so the car leans on purpose and the tire can work.

Those forces show up the instant the tire takes a side load, so roll center also changes how sharp the car feels. High can feel direct. On a high-bite lane it can feel nervous.

How you find it. From the front or the rear, draw a line through the upper link and a line through the lower arm. Where they meet is the instant center. Draw a third line from the tire contact patch through that instant center. Where that line crosses the car's centerline is the roll center.

## HOW TO FIND ROLL CENTER (front or rear view)



Track rule of thumb. Hard or slick: drop the roll center. Soft or wet: raise it so the tire can bite. At the rear, lower usually adds mid and exit grip. Too low and the car dumps and can hook-spin off the corner. After a big drop, a slightly stiffer spring or the shock moved out on the tower will often settle the chassis.

The clicks that move it the most: inner lower-arm pill height, upper-link inner ball stud, upper-link outer ball stud, and axle height in the hub (B6-era inserts). Ride height moves it too.

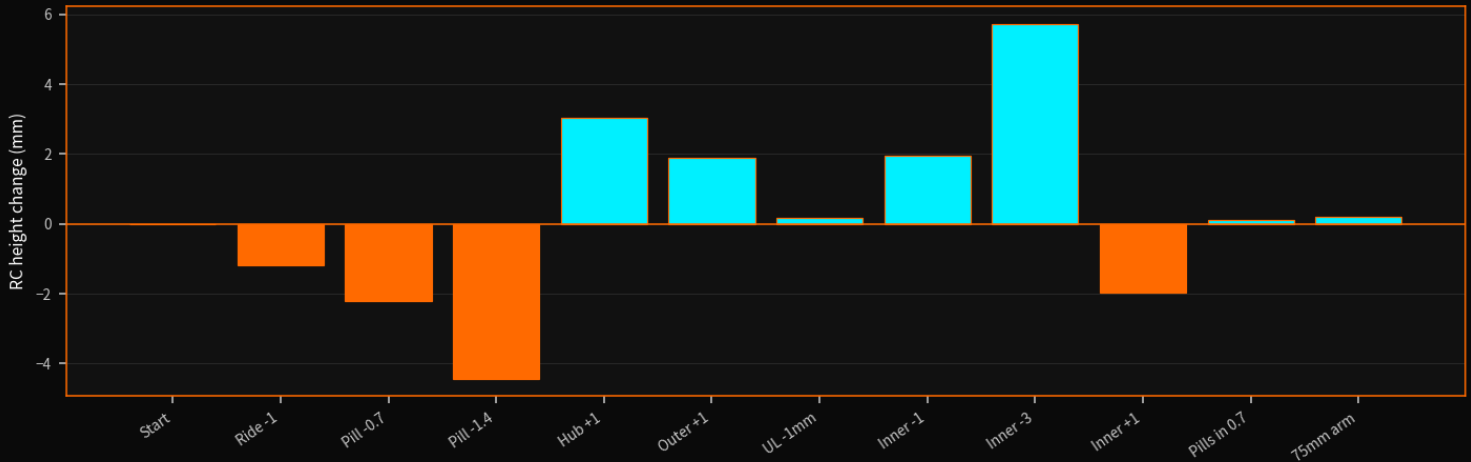
## ROUGH SENSITIVITY (ONE CLICK)

- Raise the inner pill one step (0.7 mm) → roll center up about 2.2 mm
- Raise the outer ball stud 1 mm → roll center up about 1.9 mm
- Raise the inner ball stud 1 mm → roll center DOWN about 1.9 mm
- Raise axle height one insert (0 to +1) → roll center up about 3 mm
- Drop ride height 1 mm → roll center down about 1.2 mm

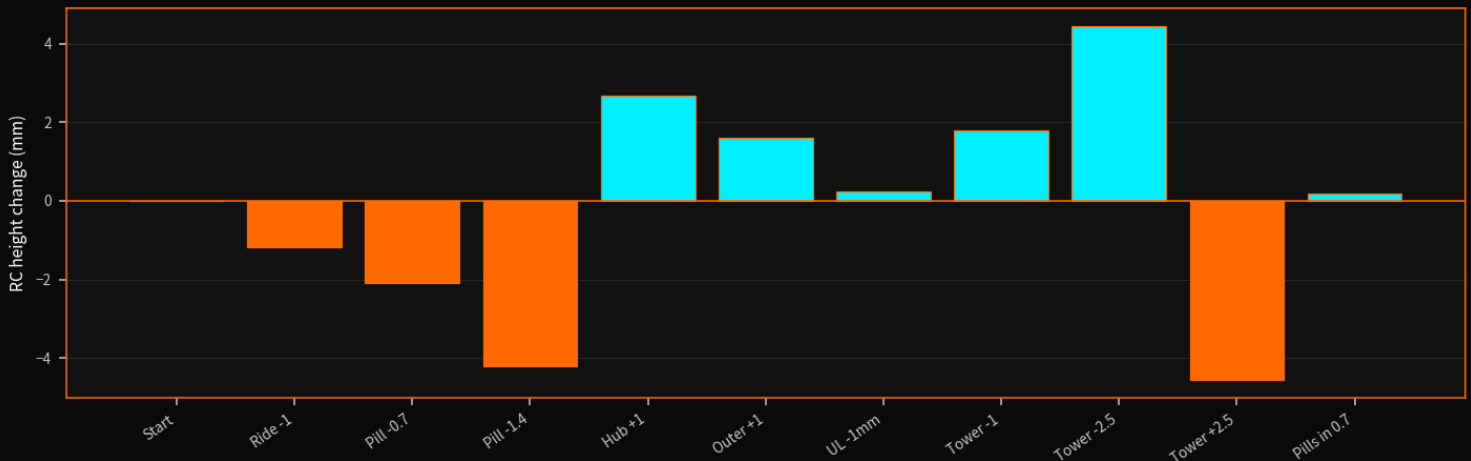
## ROLL CENTER — CHANGE PER CLICK

B6.2 START ABOUT -9.0 MM · B74 START ABOUT -7.6 MM · VALUES ARE DELTA FROM THAT START

**B6.2 REAR — start RC about 9.0 mm below ground**



**B74 REAR — start RC about 7.6 mm below ground**



Upper-link length barely moves roll center at ride height. It does change how the roll center falls as the chassis leans, and it changes camber gain. A longer upper link usually means less camber gain — the tire leans out more in roll but stays more upright in bump — and the roll center drops more as the car rolls. That can add mid-corner bite and can also hook-spin if it falls too far. Narrower pills limit how far the roll center drops in roll.

You can stack clicks to hold the same roll center and still change camber. Example: drop the pill and the inner ball stud together. Roll center stays close, camber change increases.

These plots are ride-height roll center only — what you feel at first turn-in. Camber change through the travel and roll-center migration in bump are separate jobs.

Build the car square before you chase a number. A bound hinge pin will move the chart more than a washer will.